

Exterior Porch and Stairs Estimate Geometry Worksheet

Project: Targeted Front Porch and Stair Maintenance Repaint
Property: 10375 SW Dunlin Pl, Beaverton, OR 97007
Prepared: 2026-08-24
Status: Photo-based takeoff; field verification required before scheduling.

Scope and Measurement Basis

This revision preserves physical takeoff values that reconcile to the photographs while separating geometry from coating demand. Dimensions are inferred, not field-measured. The working model is an approximately **18 ft x 5 ft porch**, a **3.5 ft-wide stair with 9 tread tops/nosings and 10 risers**, approximately **44.5 linear ft of combined porch/stair rail**, **95 balusters**, and **8 posts**.

Included gray surfaces are the porch/deck walking surface and stair tread tops/nosings. Included white surfaces are stair risers, both stair railings, the porch guardrail, and associated balusters, posts, rails, and caps.

Stringer/skirt outer faces and the deck rim band are visible in photographs, but the confirmed color scope does not establish them as white work. They are measured below for audit continuity but **excluded from coating demand and price unless authorized in writing**. Inner stringer faces, undersides, lattice, ceiling/soffit, siding, door/casing, adjacent trim, and concrete are excluded.

Audited Working Dimensions

Dimension	Working value	Confidence / field range
Porch length along house	18.0 ft	Inferred; 14-22 ft
Porch depth	5.0 ft	Inferred; 4.5-5.5 ft
Stair clear width	3.5 ft	Inferred; 3.0-4.0 ft
Stair treads	9	Photo count; verify
Stair risers	10	Derived from tread count; verify
Tread walking depth	11.25 in	Two nominal 2x6 boards plus gap
Stair rail run	10.0 ft each side	Inferred from rise/run
Porch guardrail run	24.5 ft	Inferred
Balusters	95	Inferred from 5.5 in on-center spacing
Posts	8	Inferred

The porch walking surface remains:

18.0 ft x 5.0 ft gross	= 90.00 sq ft
Less 3.5 ft x 3.0 ft stair-head opening	= 10.50 sq ft
Net porch walking surface	= 79.50 sq ft

The stair rail slope is approximately 9.78 ft from a 7.69 ft run and 6.04 ft rise; 10.0 ft per side remains a reasonable working length including terminations.

Physical Geometry Audit

Every area is a one-coat physical surface area. Excluded components remain shown so changing scope does not distort the takeoff.

ID	Component	Color status	Horizontal sq ft	Vertical sq ft	Total sq ft
A	Porch/deck walking surface	Gray, included	79.50	-	79.50

B	9 stair tread tops	Gray, included	29.53	-	29.53
C	9 tread nosing edges	Gray, included	-	3.94	3.94
D	10 stair risers	White, included	-	18.23	18.23
E	Stringer/skirt outer faces	Boundary unresolved; excluded	-	18.75	18.75
F+G	Porch and stair rail members	White, included	53.80	44.50	98.30
H	Balusters	White, included	-	106.88	106.88
I	Posts and caps	White, included	1.68	33.06	34.74
J	Deck rim/skirt band	Boundary unresolved; excluded	-	23.32	23.32
	All observed/measured components		164.51	248.68	413.19
	Included confirmed scope only		164.51	206.61	371.12

Calculation checks

- Stair tread tops: $9 \times 3.5 \text{ ft} \times 0.9375 \text{ ft} = 29.53 \text{ sq ft}$.
- Tread nosings: $9 \times 3.5 \text{ ft} \times 0.125 \text{ ft} = 3.94 \text{ sq ft}$.
- Risers: $10 \times 3.5 \text{ ft} \times 0.5208 \text{ ft} = 18.23 \text{ sq ft}$; the raw replacement riser is **1.82 sq ft** of this total.
- Combined rail: $44.5 \text{ linear ft} \times 2.209 \text{ ft audited exposed girth} = 98.30 \text{ sq ft}$.
- Balusters: $95 \times 0.500 \text{ ft girth} \times 2.25 \text{ ft exposed length} = 106.88 \text{ sq ft}$.
- Posts/caps: $26.0 \text{ linear ft of post shaft} \times 1.1667 \text{ ft girth plus } 4.40 \text{ sq ft of caps} = \mathbf{34.74 \text{ sq ft}}$.
- Excluded stringer faces: $2 \times 10.0 \text{ ft} \times 0.9375 \text{ ft} = 18.75 \text{ sq ft}$.
- Excluded rim band: $28.0 \text{ ft} \times 0.833 \text{ ft} = 23.32 \text{ sq ft}$.
- Reconciliation: included gray 112.97 + included white 258.15 = **371.12 sq ft**. Adding unresolved/excluded 42.07 returns the full observed geometry of **413.19 sq ft**.

Corrected Coating Demand

Coating demand is separate from physical geometry. "Spot repair preparation" is a condition allowance, not another coat.

Coating operation	Basis	Coverage demand
Gray localized repair/preparation	Allow approximately 20% of gray area for scraping/sanding/minor filling; actual failures only	~22.6 sq ft treated
Gray tinted primer, one complete coat	A+B+C	112.97 sq ft
Gray finish, one complete coat	A+B+C	112.97 sq ft
White localized preparation	Actual loose/failing areas, especially rail ends, cap defects, joints, and raw riser	Condition allowance; not a full-surface sand
White spot primer	Raw riser 1.82 sq ft plus exposed/repared white areas; 10% working allowance on remaining white surfaces	~27.5 sq ft
White finish, one complete coat	D+F+G+H+I	258.15 sq ft
Total liquid-coating coverage demand	Primer plus finish applications	511.59 sq ft

There is **225.94 sq ft of gray liquid-coating demand** (one full primer plus one finish) and approximately **285.65 sq ft of white liquid-coating demand** (spot primer plus one finish). This replaces the former 826.38 sq ft two-finish-coat model.

Purchase Quantity Model

Product	Theoretical demand	Planned purchase	Expected usable remainder
Tinted exterior wood/deck primer for gray surfaces	112.97 sq ft	1 gallon	Substantial remainder; retained for absorption/repair variability and touch-up stock
Gray porch-and-floor finish	112.97 sq ft	1 gallon	Substantial remainder retained and labeled
White exterior trim/rail finish	258.15 sq ft, detail-heavy	1 gallon	Moderate remainder; detail loss and touch-ups allowed
Compatible white spot primer	~27.5 sq ft	1 quart, or verified compatible stock	Limited remainder

Rated spread is not treated as purchase cost. Three full gallons plus a primer quart are required because the system uses distinct products/colors, even though theoretical use is below those container quantities. Contractor pricing and usable stock should be used where available; leftover coating remains usable project material, not waste.

Condition and Preparation Audit

- Porch floor: coating is worn and marked but substantially adhered. Wash, spot scrape/sand, minor repair, full tinted primer, and one gray finish coat.
- Stair tread tops: visibly weathered with broad coating loss. Wash and remove loose material; sand only raised, splintered, failed, or repair areas rather than restoring every tread to uniformly bare wood. Full tinted primer provides the uniform base.
- White rails: localized failure is concentrated at lower rail ends and isolated defects. Scrape/feather and spot-prime those areas; do not universally degloss sound coating unless the selected product’s adhesion test requires it.
- Risers: generally sound, with one raw replacement riser requiring primer. Minor joint repair only where paintable and non-structural.
- Split/soft/rotten wood, replacement of caps/treads/boards, structural work, and concealed deterioration remain excluded.

Field Verification and Scope Boundaries

Before purchase or scheduling, field-measure porch length/depth, stair width/count, rail footage, balusters, and posts. Confirm coating compatibility and moisture requirements. Confirm in writing that **stringer/skirt faces and the deck rim band are excluded**; add them only by written authorization after color and condition are established. Confirm the customer can provide an alternate entry during coating and cure.